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September 29, 1999

RECEIVED

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Division of Minerals & Geology

Christina L. Kamnikar
Environmental Protection Specialist
Colorado Division of Minerals and Geology
1313 Sherman St., Room 215
Denver, Colorado 80203

Subject: Report on Storage of Cement Kiln Dust at the Southdown, Inc.,
Lyons Plant, Lyons Quarry, Permit # M-77-208

Dear Christina:

Southdown has received letters from you and Harry Posey requesting clarification on certain items contained in the above referenced report. This letter contains Southdown's response (responses are keyed to the numbered items in your September 7, 1999 letter to John Lohr, Southdown, Inc.).

Item No. 1: Sources of Information on Limestone, Shale and Alluvium

The sources of information on the various geologic units present are:

- The NRCS soil survey
- USGS publications and maps on the geology of the area
- Southdown's experience over 30 years of mining the property
- Extensive drilling in the alluvium of St. Vrain Creek by gravel mining interests
- Exploratory drilling at Dowe Flats where the geologic units are the same, and
- Logs of water well drilling in the vicinity

The distances from C-Pit and the sedimentary strata to the alluvium were calculated both from maps and from field measurements. The geomorphic

expression of the alluvium is quite clear compared to the outcrops and subcrops of the sedimentary rocks.

Item No. 2: Modeling of the CKD Disposal Site as an Open vs. Closed Disposal Site.

The model simulations were indeed done under the assumption of a closed disposal site. The rationale for this was that this would be the worst case scenario, the CKD disposal site would be full and the long term, post closure conditions would have to be modeled to determine long term impacts, if any.

Even though Harry Posey has stated that conservative assumptions were used in this analysis, Southdown will commit to interim capping of the disposal site. This management practice is in addition to the other management practices described in the report referenced previously.

The "worst case" leachate volume above which potential off-site water quality impacts could begin to occur is 350,000 cubic feet per year. This volume of water would have to pass through the CKD mass and flow out of the bottom of the disposal site in order to even approach off-site impacts. To prevent this situation from happening, Southdown will place at least 2-feet of compacted shale and clay over the CKD when the disposal area reaches 7 acres in areal extent and 20 feet in thickness.

The following section of this letter discusses the results of a additional model simulations that demonstrate that this interim capping practice will never allow the leachate volume to approach the threshold volume of 350,000 cubic feet per year.

Two additional HELP model runs were made to simulate the above described interim capping of the disposed CKD. An initial run was made to simulate an open, partially-filled CKD disposal site. This initial run was simulated as a 7 acre disposal site with approximately 20 feet of CKD capped by an interim 2-foot thick compacted shale/clay cover. Since the disposal site was simulated as an open site, precipitation run-off was not allowed to occur. The default soil type used to simulate the disposed CKD (material texture 10) was the same as that used in the previously submitted HELP model. The compacted shale/clay cover was simulated as a compacted clay (material texture 25) with an effective saturated hydraulic conductivity of 3.60E-06 centimeters per second (cm/sec). Default evapotranspiration and precipitation data for Denver, Colorado were used in the simulation.

The model results of the open, partially-filled CKD disposal site indicates that the flux of leachate that will be generated at the base of the disposed CKD is about 0.3 inches per year. For this 7 acre disposal site, the annual leachate volume predicted is about 7,592 cubic feet per year. A comparison of the leachate flux generated under the interim capping scenario (0.3 inches per year) with the

leachate flux previously calculated for the closed CKD disposal site (0.74 inches per year), suggests that the interim capping scenario will reduce the leachate flux through the open disposal site by about 40 percent.

An additional HELP model run was made to simulate a closed CKD disposal site that used the interim capping approach. This additional model run was simulated as a 20-acre CKD disposal site with approximately 20 feet of CKD capped by a compacted shale/clay cover overlain by an additional 15 feet of CKD capped by a compacted shale/clay cover. The upper compacted shale/clay cap was covered with two feet of topsoil and vegetated with a poor stand of grass. The topsoil, disposed CKD, and compacted shale/clay layers were simulated using material textures 8, 10, and 25, respectively. Default evapotranspiration and precipitation data for Denver, Colorado were used in the simulation.

The model results of the closed CKD disposal site indicates that the flux of leachate that will be generated at the base of the disposed CKD is about 0.04 inches per year, or an annual leachate volume of 3,232 cubic feet. A comparison of the leachate flux under this modified closure scenario (0.04 inches per year) versus the leachate flux for the previously simulated closed CKD disposal site (0.74 inches per year), suggests that the modified closure scenario will reduce the leachate flux through the closed disposal site by about 95 percent.

The HELP model output for each of the model runs is attached.

Item No. 3: Clarification of HELP Model Data Sets.

The HELP model runs presented in previously submitted Appendix D were made using the HELP model default soil types that most closely approximated the estimated saturated hydraulic conductivities of the topsoil cover, disposed CKD, and the underlying shaley bedrock formations (i.e., Niobrara Formation, Carlile Shale, Greenhorn Limestone, Graneros Shale, and Mowry Shale). The soil type and material textures used to simulate each of these components and the effective saturated hydraulic conductivities simulated are listed in the table below.

Material (Thickness)	Default Soil Type (material texture number)	Effective Saturated Hydraulic Conductivity (cm/sec)
Topsoil Cover (2 feet)	Silt (8)	3.70E-04
Disposed CKD (35 feet)	Clayey Silt (10)	1.20E-04
Shaley Bedrock Formations (700 feet)	Compacted Clay (16)	1.00E-07

Please let me know if this letter adequately responds to your request?

Sincerely,



Paul Banks, Banks and Gesso, LLC

RICHARD HENRY

BY PB 9/30/99

Richard Henry, Grant Environmental, Inc.

PB/kj

Cc: John Lohr, Southdown

Attachments